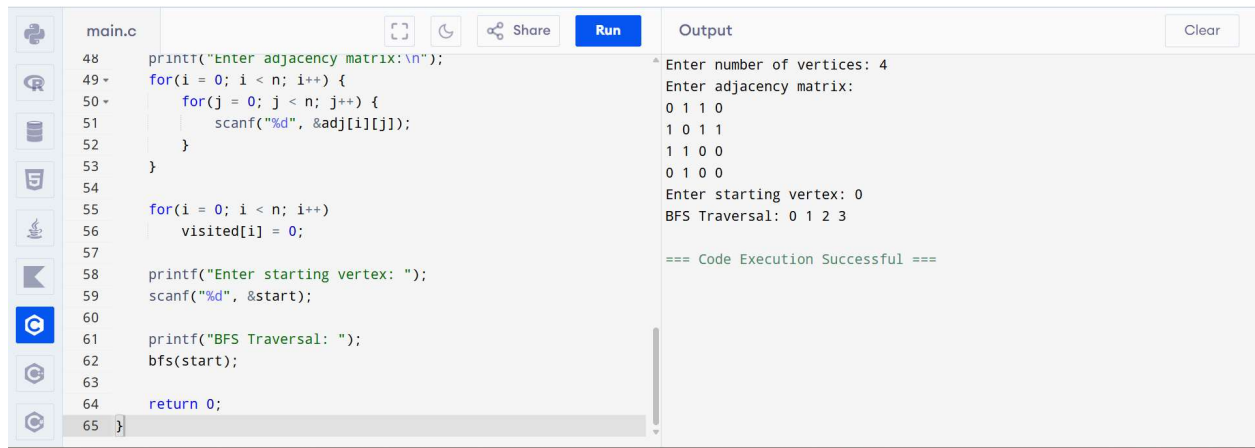




The screenshot shows a C++ IDE with a file named 'main.c'. The code implements a Breadth-First Search (BFS) algorithm. It prompts the user to enter the number of vertices (4) and the adjacency matrix. The matrix is: 0 1 1 0, 1 0 1 1, 1 1 0 0, 0 1 0 0. It then prompts for a starting vertex (0) and outputs the BFS traversal: 0 1 2 3. The output panel shows the execution was successful.

```
main.c
48 printf("Enter adjacency matrix:\n");
49 for(i = 0; i < n; i++) {
50     for(j = 0; j < n; j++) {
51         scanf("%d", &adj[i][j]);
52     }
53 }
54
55 for(i = 0; i < n; i++)
56     visited[i] = 0;
57
58 printf("Enter starting vertex: ");
59 scanf("%d", &start);
60
61 printf("BFS Traversal: ");
62 bfs(start);
63
64 return 0;
65 }
```

Output

```
Enter number of vertices: 4
Enter adjacency matrix:
0 1 1 0
1 0 1 1
1 1 0 0
0 1 0 0
Enter starting vertex: 0
BFS Traversal: 0 1 2 3

=== Code Execution Successful ===
```

Queue operations:

1. Start with node 0
Queue → [0]
2. Dequeue 0, visit it
Add neighbors → 1, 2
Queue → [1, 2]
3. Dequeue 1, visit it
Add neighbor → 3
Queue → [2, 3]
4. Dequeue 2, visit it
No new nodes
Queue → [3]
5. Dequeue 3, visit it
Queue empty